Equations of Parallel & Perpendicular Lines — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

☞ *Video: Slope rules*

Key Idea

__Parallel__ lines have the __same__ slope.

__Perpendicular__ lines have slopes that are __negative reciprocals__.

Flip the fraction **and** change the __sign__: $\frac{2}{3} \to$ $-\frac{3}{2}$, and $\frac{2}{3} \cdot \left(-\frac{3}{2}\right) =$ -1.

☞ *Video: Find the slopes*

Watch Me — Worked Example

$y = \frac{2}{3}x + 1$: slope $\frac{2}{3}$ parallel slope $\frac{2}{3}$ perpendicular slope $-\frac{3}{2}$

☞ *Video: Write a parallel line*

Watch Me — Worked Example

Through $(2, 3)$, parallel to $y = 4x - 1$.

same slope: $m =$ 4

$y - 3 = 4(x - 2)$ __point-slope__ form

$y =$ $4x - 5$

☞ *Video: Write a perpendicular line*

Watch Me — Worked Example

Through $(4, 1)$, perpendicular to $2x + 3y = 6$.

solve for y: $y = -\frac{2}{3}x + 2$ slope $-\frac{2}{3}$

perpendicular slope $\frac{3}{2}$

$y - 1 = \frac{3}{2}(x - 4)$ $y =$ $\frac{3}{2}x - 5$

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☞ *Video: Watch out*

Common Mistake

Perpendicular needs **both** steps: flip **and** change the sign. The perpendicular slope of 2 is $-\frac{1}{2}$, not $\frac{1}{2}$.

☞ *Video: Your turn*

Your Turn

Write the line through $(3, -2)$ perpendicular to $y = -3x + 7$.

Slope $\frac{1}{3}$: $y + 2 = \frac{1}{3}(x - 3) \Rightarrow y = \frac{1}{3}x - 3$

Notice

Remember: Parallel: same slope. Perpendicular: negative reciprocal — flip and change the sign. Then use point-slope form $y - y_1 = m(x - x_1)$.

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